# Gavin's Lectures on Physics - Statistical Mechanics



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# 1 Linkage : Thermodynamics Statistical Mechanics

## 1.1 Boltzmann Entropy Formula

The entropy is expressed as the logarithm of the number of microstates:

$$S = k_B \ln \Omega \tag{1}$$

## 1.2 Thermodynamic Identity

The combined first and second laws for a system with variable particle number are:

$$TdS = dQ = dU + PdV - \mu dN \tag{2}$$

The temperature, pressure and chemical potential are defined as partial derivatives of the entropy:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{V,N}, \quad \frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_{U,N}, \quad \frac{\mu}{T} = -\left(\frac{\partial S}{\partial N}\right)_{U,V} \tag{3}$$

# 2 Laws of Thermodynamics

## 2.1 Zeroth Law of Thermodynamics

Transitivity of thermal equilibrium: if two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other.

$$T_1 = T_2,\ T_3 = T_2 \implies T_1 = T_3$$

## 2.2 First Law of Thermodynamics

Law of conservation of energy. In this document, $dW$ denotes the work done *on* the system (so that for a gas, $dW = -PdV$).

$$dU = dW + dQ$$

## 2.3 Second Law of Thermodynamics

Entropy increase principle: the entropy of an isolated system never decreases, and any spontaneous process increases the total entropy.

$$dS \geq \frac{\delta Q}{T}, \quad \Delta S_{\text{isolated}} \geq 0$$

## 2.4 Third Law of Thermodynamics

Nernst's theorem: as the temperature approaches absolute zero, the entropy change of any isothermal process approaches zero.

$$\lim_{T \to 0} (\Delta S)_T = 0$$

# 3 Basic Statistical Distributions

Define:

$$\alpha = -\frac{\mu}{k_B T}, \quad \beta = \frac{1}{k_B T} \tag{4}$$

$a_l$: particle number in level $l$
$w_l$: degeneracy in level $l$
$\varepsilon_l$: energy of a single particle in level $l$
Then

$$N = \sum_l a_l, \qquad U = \sum_l a_l \varepsilon_l.$$

Using Stirling's approximation $\ln(m!) \approx m \ln m - m$ and maximizing $\ln \Omega$ subject to constraints gives

$$\Phi = \ln \Omega - \alpha \left( \sum_l a_l - N \right) - \beta \left( \sum_l a_l \varepsilon_l - E \right)$$

Number of micro states satisfies:

$$0 = \frac{\partial \Phi}{\partial a_l} = \frac{\partial \ln \Omega}{\partial a_l} - \alpha - \beta \varepsilon_l \tag{5}$$

## 3.1 Maxwell-Boltzmann Distribution (Classical)

For the classical case where $a_l \ll w_l$, the number of microstates is

$$\Omega_{\mathrm{MB}} = N! \prod_l \frac{w_l^{a_l}}{a_l!}, \quad (a_l \ll w_l).$$

Taking the logarithm and applying Stirling's approximation gives

$$\ln \Omega_{\mathrm{MB}} \approx \ln(N!) + \sum_l \left( a_l \ln w_l - a_l \ln a_l + a_l \right).$$

Differentiation with respect to $a_l$ yields

$$\alpha + \beta \varepsilon_l = \frac{\partial \ln \Omega_{\mathrm{MB}}}{\partial a_l} \approx \ln \frac{w_l}{a_l}.$$

Thus the Maxwell–Boltzmann distribution is

$$a_l = w_l e^{-\alpha - \beta \varepsilon_l} = w_l e^{-(\varepsilon_l - \mu)/k_B T} \tag{6}$$

## 3.2 Bose-Einstein Distribution (Bosons)

For bosons, the number of ways to distribute $a_l$ identical particles among $w_l$ degenerate states is given by the combinatorics of dividers:

$$\Omega_{\mathrm{BE}} = \prod_l \frac{(w_l + a_l - 1)!}{a_l! \, (w_l - 1)!}.$$

Using Stirling's approximation, the logarithm becomes

$$\ln \Omega_{\mathrm{BE}} \approx (w_l + a_l - 1)\ln(w_l + a_l - 1) - a_l \ln a_l - (w_l - 1)\ln(w_l - 1) \\ - (w_l + a_l - 1) + a_l + (w_l - 1).$$

Differentiation gives

$$\alpha + \beta\varepsilon_l = \frac{\partial \ln \Omega_{\mathrm{BE}}}{\partial a_l} \approx \ln\left(\frac{w_l}{a_l} + 1\right).$$

Hence the Bose–Einstein distribution is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} - 1} = \frac{w_l}{e^{(\varepsilon_l - \mu)/k_B T} - 1} \tag{7}$$

## 3.3 Fermi-Dirac Distribution (Fermions)

For fermions, each state can be occupied by at most one particle, so the multiplicity is

$$\Omega_{\mathrm{FD}} = \prod_l C_{w_l}^{a_l} = \prod_l \frac{w_l!}{a_l!\,(w_l - a_l)!}.$$

Using Stirling's approximation,

$$\ln \Omega_{\mathrm{FD}} \approx w_l \ln w_l - w_l - a_l \ln a_l + a_l \\ - (w_l - a_l)\ln(w_l - a_l) + (w_l - a_l).$$

Differentiation yields

$$\alpha + \beta\varepsilon_l = \frac{\partial \ln \Omega_{\mathrm{FD}}}{\partial a_l} = \ln\left(\frac{w_l}{a_l} - 1\right).$$

Thus the Fermi–Dirac distribution is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} + 1} = \frac{w_l}{e^{(\varepsilon_l - \mu)/k_B T} + 1} \tag{8}$$

In the classical limit $a_l \ll w_l$, both Bose and Fermi distributions reduce to the Maxwell–Boltzmann form, as seen from

$$\Omega_{\mathrm{BE}},\ \Omega_{\mathrm{FD}} \xrightarrow{a_l \ll w_l} \frac{\Omega_{\mathrm{MB}}}{N!}.$$

The maximisation of $\Omega(a_l)$ determines the final occupation numbers.

# 4 Statistics of Distributions

## 4.1 Boltzmann Statistics

### 4.1.1 Conservation of energy

The first law in differential form:

$$\begin{aligned} dU &= dW + dQ \\ &= Ydy + TdS, \end{aligned}$$

and also

$$\sum_l d(a_l\varepsilon_l) = \sum_l a_l d\varepsilon_l + \sum_l \varepsilon_l da_l.$$

The connection between macroscopic and microscopic world:

$$\begin{aligned} dU &= \sum_l d(a_l\varepsilon_l) \\ dW &= Ydy = \sum_l a_l d\varepsilon_l \\ dQ &= TdS = \sum_l \varepsilon_l da_l \end{aligned} \tag{9}$$

### 4.1.2 Thermodynamic quantity

For Boltzmann statistics, the occupation numbers and the entropy logarithm are:

$$\begin{cases} a_l = w_l e^{-\alpha-\beta\varepsilon_l} \\ \ln\Omega_{\mathrm{MB}} \approx N\ln N + \alpha N + \beta U \end{cases}$$

We define the single-particle partition function

$$Z_1 = \sum_l w_l e^{-\beta\varepsilon_l} \tag{10}$$

Then the total particle number and energy are expressed as

$$\begin{cases} N = \sum_l a_l = e^{-\alpha} Z_1 \\ U = \sum_l a_l\varepsilon_l = -N\frac{\partial \ln Z_1}{\partial\beta} \end{cases} \tag{11}$$

As $\omega_l$ and $\beta$ are independent of y, only $\varepsilon_l$ is a function of y. Then,

$$Z_1 = Z_1(y,\beta)$$

The generalised force conjugate to an external parameter $y$ is

$$Y = \sum_l \frac{\partial\varepsilon_l}{\partial y} a_l = -\frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y},$$

so the work done on the system is

$$dW = Ydy = -\frac{N}{\beta}\left(d\ln Z_1 - \frac{\partial \ln Z_1}{\partial\beta} d\beta\right).$$

From the first law, we obtain

$$\begin{aligned} TdS = dQ &= dU - dW \\ &= -Nd\frac{\partial \ln Z_1}{\partial \beta} + \frac{N}{\beta}\left(d\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}d\beta\right) \\ &= \frac{N}{\beta}d\left[\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}\beta\right]. \end{aligned}$$

Setting $\beta = 1/(k_B T)$, we get

$$\begin{aligned} S &= k_B N\left(\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}\beta\right) \\ &= k_B N\left(\ln N + \alpha + \beta\frac{U}{N}\right) \\ &= k_B\left(N\ln N + \alpha N + \beta U\right). \end{aligned}$$

Thus the entropy is given by

$$S = k_B \ln \Omega_{\mathrm{MB}} \tag{12}$$

From the relations $U$ and $S$ as above:

$$\begin{cases} U = -N\frac{\partial \ln Z_1}{\partial \beta} \\ S = k_B N\left(\ln Z_1 - \frac{\partial \ln Z_1}{\partial \beta}\beta\right) \\ \beta = \frac{1}{k_B T} \end{cases}$$

The Helmholtz free energy:

$$F = U - TS = -Nk_B T \ln Z_1 \tag{13}$$

From the derivative of $\ln Z_1$ with respect to volume:

$$\begin{cases} Y = -\frac{N}{\beta}\frac{\partial \ln Z_1}{\partial y} \\ dW = Ydy = -PdV \end{cases}$$

The pressure:

$$P = \frac{N}{\beta}\frac{\partial \ln Z_1}{\partial V} \tag{14}$$

### 4.1.3 Ideal gas equation of state

For a single particle in a box, the energy and phase-space element are

$$\begin{cases} \varepsilon = \left(p_x^2 + p_y^2 + p_z^2\right)/(2m) \\ dw = dxdydz\ dp_x dp_y dp_z/h^3 \end{cases}$$

Then

$$Z_1 = \sum_l w_l e^{-\beta\varepsilon_l} = \int e^{-\beta\varepsilon} dw = V\left(\frac{2\pi m}{h^2\beta}\right)^{\frac{3}{2}}.$$

Using the pressure formula above, we find

$$P = \frac{N}{\beta}\frac{\partial \ln Z_1(\beta, V)}{\partial V} = \frac{N}{\beta V} = \frac{N k_B T}{V},$$

which yields the ideal gas law:

$$PV = N k_B T \tag{15}$$

The condition for the classical (ideal) gas is

$$e^{\alpha} = \frac{Z_1}{N} = \frac{V}{N}\left(\frac{2\pi m}{h^2 \beta}\right)^{\frac{3}{2}} \gg 1, \quad E_p = 0,\ \varepsilon = \frac{1}{2}mv^2.$$

### 4.1.4 Maxwell velocity distribution Law

Starting from the phase-space element

$$dw = \frac{dxdydz\ dp_x dp_y dp_z}{h^3},$$

we have

$$\begin{aligned} Z_1 &= \int e^{-\beta\varepsilon} dw = \frac{V}{h^3}\int e^{-\frac{mv^2}{2k_B T}} m^3 dv_x dv_y dv_z \\ &= \frac{m^3 V}{h^3}\int_0^{\infty} e^{-\frac{mv^2}{2k_B T}} \cdot 4\pi v^2 dv. \end{aligned}$$

Using derived partition function,

$$Z_1 = V\left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}$$

We get the Maxwell speed distribution:

$$f(v) = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-mv^2/(2k_B T)} \tag{16}$$

With normalization:

$$\int_0^{\infty} f(v)dv = 1$$

### 4.1.5 Velocities

The Gamma function is

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt,$$

with properties $\Gamma(z+1) = z\,\Gamma(z), \quad \Gamma(n) = (n-1)!, \quad \Gamma(\frac{1}{2}) = \sqrt{\pi}$.

The most probable, mean, and root-mean-square speeds are

$$\begin{cases} f'(v_p) = 0 \implies v_p = \sqrt{\frac{2k_B T}{m}} \\ \langle v \rangle = \int_0^{\infty} v f(v) dv = \sqrt{\frac{8k_B T}{\pi m}} \\ \sqrt{\langle v^2 \rangle} = \left[\int_0^{\infty} v^2 f(v) dv\right]^{\frac{1}{2}} = \sqrt{\frac{3k_B T}{m}}. \end{cases}$$

### 4.1.6 The Equipartition Theorem

For a single-atom molecule with 3 degrees of freedom.

Consider partition,

$$Z_1 = V\left(\frac{2\pi m k_B T}{h^2}\right)^{\frac{3}{2}} = V\left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3}{2}}$$

Total internal energy,

$$U = -N\frac{\partial \ln Z_1}{\partial \beta} = \frac{3}{2}\frac{N}{\beta} = \frac{3}{2}Nk_BT$$

Then the average energy for the $x$-direction is

$$\langle \varepsilon_x \rangle = \frac{1}{3}\frac{U}{N} = \frac{1}{2}k_BT.$$

Each degree of freedom contributes $\frac{1}{2}k_BT$ to the average energy.

In general, the total internal energy is

$$U = \frac{s}{2}Nk_BT, \tag{17}$$

where $s$ is the number of degrees of freedom.

### 4.1.7 Heat Capacity

The heat capacities,

$$\begin{cases} C_V = \frac{\partial U}{\partial T}\big|_V \\ C_P = \frac{\partial H}{\partial T}\big|_P \end{cases}$$

From

$$\begin{cases} PV = Nk_BT \\ H = U + PV \end{cases}$$

We get the relation of capacities,

$$C_P - C_V = Nk_B \tag{18}$$

Define the capacity ratio:

$$\gamma = \frac{C_P}{C_V}$$

For a monatomic gas (He, Ne, Ar):
$s = 3,\ U = \frac{3}{2}Nk_BT,\ C_V = \frac{3}{2}Nk_B,\ C_P = \frac{5}{2}Nk_B,\ \gamma = \frac{5}{3}$.
For a diatomic gas ($H_2, O_2, \ldots$):
$s = 5,\ U = \frac{5}{2}Nk_BT,\ C_V = \frac{5}{2}Nk_B,\ C_P = \frac{7}{2}Nk_B,\ \gamma = \frac{7}{5}$.

## 4.2 Bose Statistics

For bosons, the occupation number is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} - 1}.$$

The grand partition function,

$$\mathcal{Z} = \prod_l \left(1 - e^{-\alpha-\beta\varepsilon_l}\right)^{-w_l} \tag{19}$$

So,

$$\ln \mathcal{Z} = -\sum_l w_l \ln\left(1 - e^{-\alpha-\beta\varepsilon_l}\right).$$

The total particle number and energy,

$$\begin{cases} N = \sum_l a_l = -\frac{\partial \ln \mathcal{Z}}{\partial \alpha} \\ U = \sum_l a_l \varepsilon_l = -\frac{\partial \ln \mathcal{Z}}{\partial \beta}. \end{cases} \tag{20}$$

With $dW = Ydy = -PdV$, the generalised force is

$$Y = \sum_l \frac{\partial \varepsilon_l}{\partial y} a_l = -\frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial y},$$

and the pressure is

$$P = \frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial V},$$

## 4.3 Fermi Statistics

For fermions, the occupation number is

$$a_l = \frac{w_l}{e^{\alpha+\beta\varepsilon_l} + 1}.$$

The grand partition function,

$$\mathcal{Z} = \prod_l \left(1 + e^{-\alpha-\beta\varepsilon_l}\right)^{w_l} \tag{21}$$

So,

$$\ln \mathcal{Z} = \sum_l w_l \ln\left(1 + e^{-\alpha-\beta\varepsilon_l}\right).$$

The total particle number and energy,

$$\begin{cases} N = \sum_l a_l = -\frac{\partial \ln \mathcal{Z}}{\partial \alpha} \\ U = \sum_l a_l \varepsilon_l = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} \end{cases}$$

With $dW = Ydy = -PdV$, the generalised force is

$$Y = \sum_l \frac{\partial \varepsilon_l}{\partial y} a_l = -\frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial y},$$

and the pressure is

$$P = \frac{1}{\beta}\frac{\partial \ln \mathcal{Z}}{\partial V},$$

# 5 Ensembles' Theory

## 5.1 Definition of Ensemble

An ensemble is a collection of identical copies of a system, each representing a possible microstate of the real system.

As $N \to \infty$, all ensembles are equivalent for large systems, differing only in their fixed macroscopic variables.

## 5.2 The Three Ensembles

### 5.2.1 Microcanonical Ensemble

Fixed: $N, V, E$.

Potential: $S = k_B \ln \Omega$.

Example: Einstein Solid Multiplicity. For $N$ independent quantum harmonic oscillators and $q$ identical energy quanta, the multiplicity is

$$\Omega(N, q) = \frac{(q + N - 1)!}{q!\,(N - 1)!}.$$

### 5.2.2 Canonical Ensemble

Fixed: $N, V, T$.

Potential: $F = U - TS = -k_B T \ln Z$.

The canonical partition function is the sum over microstates with the Boltzmann factor:

$$Z = \sum_S e^{-\beta E_S}, \qquad \beta = \frac{1}{k_B T}. \tag{22}$$

The internal energy is obtained from the derivative of $\ln Z$ with respect to $\beta$:

$$U = \overline{E} = -\frac{\partial \ln Z}{\partial \beta}. \tag{23}$$

The entropy in terms of the partition function is

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N} = k_B \left[\ln Z + T \frac{\partial \ln Z}{\partial T}\right]. \tag{24}$$

The chemical potential follows from the dependence of $F$ on $N$:

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} = -k_B T \left(\frac{\partial \ln Z}{\partial N}\right)_{T,V}. \tag{25}$$

The heat capacity is related to energy fluctuations:

$$C_V = \left(\frac{\partial \overline{E}}{\partial T}\right)_V = \frac{1}{k_B T^2} \sigma_E^2, \qquad \sigma_E^2 = \overline{E^2} - (\overline{E})^2.$$

### 5.2.3 Grand Canonical Ensemble

Fixed: $\mu, V, T$.

Potential: $\Omega_G = F - \mu N = -k_B T \ln \mathcal{Z}$.

The grand partition function sums over all particle numbers and microstates:

$$\mathcal{Z} = \sum_{N=0}^{\infty} \sum_{S} e^{-\beta(E_S - \mu N_S)}. \tag{26}$$

The average particle number is obtained from the grand potential:

$$\overline{N} = -\left(\frac{\partial \Omega_G}{\partial \mu}\right)_{T,V}. \tag{27}$$

The internal energy from the grand partition function is

$$\overline{E} = -\left(\frac{\partial \ln \mathcal{Z}}{\partial \beta}\right)_{\mu,V} + \mu \overline{N}. \tag{28}$$

The entropy is given by the temperature derivative of the grand potential:

$$S = -\left(\frac{\partial \Omega_G}{\partial T}\right)_{\mu,V}. \tag{29}$$

## 5.3 Thermodynamic Potentials

The thermodynamic potentials are defined as:

- Internal energy: $U$
- Enthalpy: $H = U + PV$
- Helmholtz free energy: $F = U - TS$
- Gibbs free energy: $G = H - TS = F + PV$
- Grand potential: $\Omega_G = F - \mu N$

Their differential forms are:

$$\begin{cases} dU = T\,dS - P\,dV + \mu\,dN, \\ dH = T\,dS + V\,dP + \mu\,dN, \\ dF = -S\,dT - P\,dV + \mu\,dN, \\ dG = -S\,dT + V\,dP + \mu\,dN, \\ d\Omega_G = -S\,dT - P\,dV - N\,d\mu. \end{cases} \tag{30}$$

## 5.4 Maxwell Relations

From the equality of mixed second derivatives, we obtain the Maxwell relations:

$$\begin{cases} \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V, \\ \left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P, \\ \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V, \\ \left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P. \end{cases} \tag{31}$$

# 6 Applications

## 6.1 Blackbody Radiation

Ensemble : Grand Canonical with $\mu_\gamma = 0$. Distribution : Bose-Einstein gives the Planck distribution $\overline{n}_\gamma = \frac{1}{e^{\varepsilon/k_B T}-1}$.

### 6.1.1 Photon Density of States

For photons in a box, with $k_i = n_i\pi/L$, volume $V = L^3$, two polarisations and positive octant (since $n_i > 0$), the number of states in $[k, k+dk]$ is

$$g(k)\,dk = 2 \cdot \frac{1}{8} \cdot \frac{4\pi k^2 dk}{(\pi/L)^3} = \frac{V}{\pi^2} k^2 dk.$$

Using the photon dispersion relation $\varepsilon = h\nu = hck/(2\pi)$, we convert to energy space:

$$g(\varepsilon)\,d\varepsilon = g(k)\frac{dk}{d\varepsilon}\,d\varepsilon = \frac{8\pi V}{(hc)^3}\varepsilon^2 d\varepsilon.$$

The photon density of states per unit energy is therefore:

$$g(\varepsilon) = \frac{8\pi V}{(hc)^3}\varepsilon^2. \tag{32}$$

The two polarisations correspond to $S_\gamma = 1$ with $m_\gamma = \pm 1$ (the longitudinal mode $m_\gamma = 0$ is forbidden), giving linear $x/y$ or right/left circular polarisations.

### 6.1.2 Planck Energy Spectrum

The spectral energy density is obtained by multiplying three factors:

- the single-photon energy $\varepsilon$,
- the density of states $g(\varepsilon) = 8\pi V\varepsilon^2/(hc)^3$,
- the average occupation number of each energy state $\overline{n}_\gamma = 1/(e^{\varepsilon/k_B T} - 1)$ , from Bose-Einstein statistics with zero chemical potential ($\mu_\gamma = 0$).

Thus the energy per unit volume in modes with energies between $\varepsilon$ and $\varepsilon + d\varepsilon$ is

$$\begin{aligned} u(\varepsilon)\,d\varepsilon &= \frac{1}{V}\,\varepsilon\,\overline{n}_\gamma\,g(\varepsilon)\,d\varepsilon \\ &= \frac{8\pi\varepsilon^3}{(hc)^3}\frac{1}{e^{\varepsilon/k_B T}-1}\,d\varepsilon. \end{aligned}$$

Hence the Planck spectral energy density is

$$u(\varepsilon) = \frac{8\pi\varepsilon^3}{(hc)^3}\frac{1}{e^{\varepsilon/k_B T}-1}. \tag{33}$$

This spectrum correctly reproduces the Rayleigh-Jeans law in the low-frequency limit ($\varepsilon \ll k_B T$, where the exponential approximates to $1 + \varepsilon/k_B T$, giving $u(\varepsilon) \propto \varepsilon^2 k_B T$) and the Wien law in the high-frequency limit ($\varepsilon \gg k_B T$, where the exponential dominates). It resolves the ultraviolet catastrophe and was the foundation for the birth of quantum mechanics.

### 6.1.3 Stefan-Boltzmann Law

For a general electromagnetic field, the local energy density and the energy flux density are

$$w = \frac{1}{2}\left(\varepsilon E^2 + \frac{1}{\mu}B^2\right), \qquad \mathbf{S} = \frac{1}{\mu}\mathbf{E}\times\mathbf{B}.$$

The total energy in a volume and the power flowing through a surface are obtained by integration:

$$E_{\text{total}} = \int w\, dV, \qquad P = \int \mathbf{S}\cdot d\mathbf{A}.$$

For blackbody radiation inside a cavity, $w$ and $\mathbf{S}$ fluctuate rapidly in time. We are interested in their statistical averages. The average energy density is

$$\langle w\rangle = \frac{U}{V}.$$

This quantity is obtained by integrating the Planck spectrum over all photon energies:

$$\begin{aligned}\frac{U}{V} &= \int_0^\infty u(\varepsilon)\, d\varepsilon \\ &= \frac{8\pi}{(hc)^3}\int_0^\infty \frac{\varepsilon^3}{e^{\varepsilon/k_BT}-1}\, d\varepsilon \\ &= \frac{8\pi(k_BT)^4}{(hc)^3}\int_0^\infty \frac{x^3}{e^x-1}\, dx.\end{aligned}$$

Using the standard integral $\int_0^\infty x^3/(e^x-1)\, dx = \pi^4/15$, (from the Riemann zeta function $\zeta(4) = \pi^4/90$), we obtain

$$\frac{U}{V} = \frac{8\pi(k_BT)^4}{(hc)^3}\cdot\frac{\pi^4}{15} = a_S T^4, \tag{34}$$

with the radiation constant

$$a_S = \frac{8\pi^5 k_B^4}{15(hc)^3}.$$

Next we determine the average power radiated through a surface. For isotropic radiation, the flux through a surface element is reduced by the factor $\cos\theta$, where $\theta$ is the angle between the propagation direction and the surface normal. Averaging over the forward hemisphere gives

$$\langle S_n\rangle = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi/2}\frac{U}{V}\, c\cos\theta\, \sin\theta\, d\theta\, d\phi = \frac{c}{4}\frac{U}{V}.$$

Thus the power per unit area escaping through the surface is $\frac{c}{4}\frac{U}{V}$.

Finally, the total radiated power from a blackbody surface of area $A$ follows from $P = \int \mathbf{S}\cdot d\mathbf{A}$:

$$P_r = \langle S_n\rangle A = \frac{c}{4}\frac{U}{V}A.$$

Substituting $U/V = a_S T^4$ gives

$$P_r = \frac{c}{4}a_S T^4 A = \sigma T^4 A, \tag{35}$$

where the Stefan–Boltzmann constant is

$$\sigma = \frac{c}{4}a_S = \frac{2\pi^5 k_B^4}{15h^3c^2} = 5.67\times10^{-8}\ \text{W}/(\text{m}^2\cdot\text{K}^4).$$

### 6.1.4 Wien's Displacement Law

The peak of the Planck spectrum is found by maximising $u(\varepsilon)$ with respect to $\varepsilon$, i.e. solving $du/d\varepsilon = 0$. Let $x = \varepsilon/(k_B T)$. The condition $du/d\varepsilon = 0$ is equivalent to

$$\frac{d}{dx}\left(\frac{x^3}{e^x - 1}\right) = 0.$$

The positive solution of this transcendental equation is

$$x = 2.82.$$

Thus the peak energy is

$$\varepsilon_{\text{peak}} = 2.82\, k_B T.$$

Using the relation $\varepsilon = hc/\lambda$, we obtain Wien's displacement law:

$$\lambda_{\text{peak}} T = \frac{hcT}{\varepsilon_{\text{peak}}} = \frac{hc}{2.82\, k_B} = 2.898 \times 10^{-3}\ \text{m} \cdot \text{K}. \tag{36}$$

## 6.2 Black Hole Thermodynamics

### 6.2.1 Schwarzschild Radius

For a non-rotating black hole of mass $M$, the Schwarzschild radius (the event horizon radius) is

$$R_s = \frac{2GM}{c^2}.$$

The surface area of the event horizon is therefore

$$A = 4\pi R_s^2 = \frac{16\pi G^2 M^2}{c^4}.$$

### 6.2.2 Bekenstein–Hawking Entropy

We use a characteristic length scale $R_{\text{BH}} = GM/c^2$ (which is $R_s/2$) as an order-of-magnitude estimate for the typical wavelength of Hawking quanta. Note that this is not the general Kerr radius; it is a convenient scale for this heuristic estimate:

$$R_{\text{BH}} = \frac{GM}{c^2}.$$

The internal energy of the black hole is simply its rest mass energy:

$$U = Mc^2.$$

A Hawking photon has wavelength of order $R_{\text{BH}}$, so its energy is

$$\varepsilon = \frac{hc}{\lambda}, \qquad \lambda = R_{\text{BH}} = \frac{GM}{c^2}.$$

Thus $\varepsilon \sim hc^3/(GM)$.

The number of such photons that could account for the black hole's internal energy is

$$N = \frac{U}{\varepsilon} = \frac{Mc^2}{hc^3/(GM)} = \frac{GM^2}{hc}.$$

Assuming each photon contributes one bit of entropy, the entropy scales as

$$S_{\rm BH} \sim N k_B = \frac{GM^2 k_B}{hc}.$$

In terms of the horizon area $A = 16\pi G^2 M^2/c^4$, a full general-relativistic calculation gives the exact result of the Bekenstein–Hawking entropy is

$$S_{\rm BH} = \frac{8\pi^2 GM^2 k_B}{hc} = k_B \frac{\pi c^3}{2hG} A. \tag{37}$$

### 6.2.3 Hawking Temperature

The thermodynamic identity for the black hole is

$$T\, dS_{\rm BH} = dQ = dU,$$

where $U = Mc^2$. Using the exact entropy $S_{\rm BH} = 8\pi^2 GM^2 k_B/(hc)$. Thus

$$\frac{1}{T} = \left(\frac{\partial S_{\rm BH}}{\partial U}\right) = \frac{1}{c^2}\frac{dS_{\rm BH}}{dM}. = \frac{16\pi^2 GMk_B}{hc^3}.$$

Rearranging gives the Hawking temperature:

$$T_{\rm BH} = \frac{hc^3}{16\pi^2 GMk_B}. \tag{38}$$

This shows that $T_{\rm BH} \propto 1/M$: smaller black holes are hotter.

### 6.2.4 Hawking Radiation

A black hole radiates as a black body at temperature $T_{\rm BH}$. From Wien's law, the peak photon energy is $\varepsilon_{\rm peak} = 2.82\, k_B T_{\rm BH}$.

The horizon area scales as

$$A = \frac{16\pi G^2 M^2}{c^4} \propto M^2.$$

The radiated power is given by the Stefan–Boltzmann law:

$$P_r = \sigma A T_{\rm BH}^4.$$

Substituting the expressions for $A$ and $T_{\rm BH}$,using $\sigma = 2\pi^5 k_B^4/(15h^3c^2)$:

$$\begin{aligned} P_r &= \frac{2\pi^5 k_B^4}{15h^3c^2} \cdot \frac{16\pi G^2 M^2}{c^4} \cdot \left(\frac{hc^3}{16\pi^2 GMk_B}\right)^4 \\ &= \frac{c^2 \mathcal{H}}{M^2}. \end{aligned}$$

where we have defined the Hawking constant

$$\mathcal{H} = \frac{hc^4}{30720\pi^2 G^2} = 4.0 \times 10^{15}\ \text{kg}^3/\text{s}. \tag{39}$$

Energy conservation requires that the mass loss rate equals the radiated power:

$$\frac{d(Mc^2)}{dt} = -P_r = -\frac{c^2 \mathcal{H}}{M^2},$$

or equivalently

$$\frac{dM}{dt} = -\frac{\mathcal{H}}{M^2}. \tag{40}$$

### 6.2.5 Evaporation Lifetime

From the mass loss rate

$$\frac{dM}{dt} = -\frac{\mathcal{H}}{M^2},$$

so the evaporation lifetime is

$$\tau = \int_0^\tau dt = -\frac{1}{\mathcal{H}} \int_{M_{\rm BH}}^0 M^2 \, dM = \frac{M_{\rm BH}^3}{3\mathcal{H}}. \tag{41}$$

## 6.3 Degenerate Fermi Gases

Ensemble : Grand Canonical ($T \ll T_F$, fully degenerate).

Distribution : Fermi-Dirac : $\overline{n}_{\rm FD} = \dfrac{1}{e^{(\varepsilon-\mu)/k_B T}+1}$.

This section uses the quantum mechanics of free electrons in a 3D infinite square well of side $L$. States are labelled by $(n_x, n_y, n_z)$.

The longest wavelength is set by the box: for the ground state, $L = \lambda/2$, so $\lambda_{\rm max} = 2L$. The shortest wavelength is set by the highest occupied level $n_{\rm max}$, which is fixed by the electron number $N$.

Thus the Fermi energy depends on the density $N/V$, not on an external energy input.

### 6.3.1 Fermi Energy

At zero temperature, the Fermi-Dirac distribution becomes a step function:

$$\lim_{T\to 0} \overline{n}_{\rm FD} = \begin{cases} 1, & \varepsilon < \mu = \varepsilon_F, \\ 0, & \varepsilon > \mu = \varepsilon_F. \end{cases}$$

Thus all states with energy below the Fermi energy $\varepsilon_F$ are occupied, and all states above are empty.

For electrons in a cubic box of volume $V = L^3$, the quantum numbers satisfy

$$n^2 = n_x^2 + n_y^2 + n_z^2 \le n_{\rm max}^2,$$

with $n_x, n_y, n_z$ positive integers. The de Broglie wavelength and momentum are

$$\lambda_n = \frac{2L}{n}, \qquad p_n = \frac{h}{\lambda_n} = \frac{hn}{2L} = \frac{hn}{2V^{1/3}}.$$

Counting both spin states (factor of 2) and integrating over the positive octant (factor of 1/8), the total number of electrons is

$$N = 2 \times \frac{1}{8} \times \frac{4\pi}{3} n_{\rm max}^3 = \frac{\pi}{3} n_{\rm max}^3.$$

**Non-relativistic case** ($v \ll c$)**:** The non-relativistic single-particle energy is therefore

$$\varepsilon_n = \frac{p_n^2}{2m_e} = \frac{h^2 n^2}{8 m_e V^{2/3}}.$$

The Fermi energy: the energy at the Fermi surface.

$$\varepsilon_F = \varepsilon_{\max} = \frac{h^2 n_{\max}^2}{8 m_e V^{2/3}} = \frac{h^2}{8 m_e}\left(\frac{3N}{\pi V}\right)^{2/3} = k_B T_F, \tag{42}$$

where $T_F$ is the Fermi temperature. For constant particle number N, we have $\varepsilon_F \propto V^{-2/3}$.

The total internal kinetic energy at zero temperature is obtained by summing over all occupied states:

$$\begin{aligned} U_k &= 2 \sum_{n_x, n_y, n_z \geq 1} \varepsilon(n_x, n_y, n_z) \\ &= 2 \times \frac{1}{8} \int_0^{n_{\max}} \varepsilon_n \cdot 4\pi n^2 \, dn \\ &= \frac{\pi h^2}{40 m_e V^{2/3}} n_{\max}^5 \\ &= \frac{\pi h^2}{40 m_e V^{2/3}} (\frac{3N}{\pi})^{5/3} \\ &= \frac{3}{5} N \varepsilon_F \end{aligned}$$

Thus the internal energy scales with volume as $U \propto \varepsilon_F \propto V^{-2/3}$.

Neglecting gravitational energy, so that $U = U_k$.

The degeneracy pressure is obtained from the thermodynamic relation $P_e = -(\partial U/\partial V)_{S,N}$.

$$P_e = -\left(\frac{\partial U}{\partial V}\right)_{S,N} = \frac{2}{3}\frac{U}{V} = \frac{2}{5}\frac{N}{V}\varepsilon_F. \tag{43}$$

This shows that $P_e \propto \varepsilon_F / V \propto V^{-5/3}$.

The bulk modulus (体模量) is

$$B = -V\left(\frac{\partial P_e}{\partial V}\right)_T = \frac{5}{3} P_e = \frac{10}{9}\frac{U}{V}. \tag{44}$$

**Relativistic case** ($v \sim c$)**:** The relativistic single-particle energy is therefore

$$\varepsilon_n \approx p_n c = \frac{hcn}{2V^{1/3}}.$$

The Fermi energy: the energy at the Fermi surface.

$$\varepsilon_F = \varepsilon_{\max} = \frac{hcn_{\max}}{2V^{1/3}} \tag{45}$$

For constant particle number N, we have $\varepsilon_F \propto V^{-1/3}$.

The total internal kinetic energy at zero temperature is obtained by summing over all occupied states:

$$\begin{aligned} U_k &= 2 \times \frac{1}{8} \int_0^{n_{\max}} \varepsilon_n \cdot 4\pi n^2 \, dn \\ &= \frac{\pi hc}{8V^{1/3}} n_{\max}^4 \\ &= \frac{\pi hc}{8V^{1/3}} (\frac{3N}{\pi})^{4/3} \\ &= \frac{3}{4} N \varepsilon_F \end{aligned}$$

Thus the internal energy scales with volume as $U \propto \varepsilon_F \propto V^{-1/3}$.

### 6.3.2 White Dwarf Stars

In the fully degenerate limit ($T \approx 0$), a white dwarf star is supported against gravitational collapse by electron degeneracy pressure.

For a spherical star,

$$V = \frac{4}{3}\pi R^3$$

Assuming uniform density, the mass inside radius r is

$$m(r) = \frac{4\pi}{3} r^3 \rho$$

The mass of the white dwarf star

$$M_{\rm WD} = \frac{4\pi}{3} R^3 \rho,$$

where $\rho$ is the mass density.

For a uniform sphere, the gravitational potential energy is

$$U_p = -\int_0^R \frac{Gm(r)}{r}\, dm = -\frac{3}{5}\frac{GM_{\rm WD}^2}{R} = -\frac{\alpha_p}{R},$$

with

$$\alpha_p = \frac{3}{5} G M_{\rm WD}^2.$$

Assuming the white dwarf is composed of fully ionised helium (or carbon/oxygen), each electron is associated with two nucleons ($\mu_e = A/Z = 2$), so

$$M_{\rm WD} = \mu_e N m_p = 2 N m_p,$$

**Non-relativistic case** ($v \ll c$)**:** For non-relativistic electrons, the kinetic energy can be written as

$$U_k = \frac{\pi h^2}{40 m_e V^{2/3}} (\frac{3N}{\pi})^{5/3} = \frac{\beta_k}{R^2}.$$

where

$$\beta_k = \frac{3}{40}\left(\frac{9}{4\pi^2}\right)^{2/3} \frac{h^2}{m_e}\left(\frac{M_{\rm WD}}{2m_p}\right)^{5/3}.$$

The equilibrium condition is obtained by minimizing the total energy $U_k + U_p$ with respect to $R$:

$$\frac{d}{dR}(U_k + U_p) = 0.$$

Since $U_k = \beta_k R^{-2}$ and $U_p = -\alpha_p R^{-1}$, we have

$$-\frac{2\beta_k}{R^3} + \frac{\alpha_p}{R^2} = 0.$$

Thus the virial theorem gives

$$2U_k + U_p = 0.$$

Thus:

$$R = \frac{2\beta_k}{\alpha_p} = 0.029 \frac{h^2}{G m_e m_p^{5/3} M_{\rm WD}^{1/3}}. \tag{46}$$

**Relativistic case** ($v \sim c$)**:** The kinetic energy in the relativistic limit is

Using the same $V$ and $N$ relations as above, we obtain

$$\begin{aligned} U_k &= \frac{\pi hc}{8V^{1/3}}(\frac{3N}{\pi})^{4/3} \\ &= \frac{\pi}{8}\left(\frac{3}{2\pi}\right)^{4/3}\left(\frac{3}{4\pi}\right)^{1/3}\left(\frac{M_{\rm WD}}{m_p}\right)^{4/3}\frac{hc}{R} \\ &\approx 0.091\left(\frac{M_{\rm WD}}{m_p}\right)^{4/3}\frac{hc}{R}. \end{aligned}$$

The gravitational potential energy remains

$$U_p = -\frac{3}{5}\frac{GM_{\rm WD}^2}{R}.$$

The equilibrium condition is obtained by minimizing the total energy $U_k + U_p$ with respect to $R$:

$$\frac{d}{dR}(U_k + U_p) = 0.$$

We get the virial theorem in relativistic limit as $U_k + U_p = 0$.

The Chandrasekhar mass limit:

$$M_{\rm WD} \leq \left(\frac{5}{3}\times 0.091 \times \frac{hc}{G}\right)^{3/2}\frac{1}{m_p^2} \approx 1.7M_\odot.$$

The full general-relativistic calculation gives a slightly lower value:

$$M_{\rm WD} \leq 1.44M_\odot. \tag{47}$$

### 6.3.3 Neutron Stars

For neutron stars, the degeneracy pressure comes from neutrons rather than electrons. Neutrons have much larger mass, consider the case in the non-relativistic regime, we have

$$U_k = \frac{3}{5}N\varepsilon_F, \quad \varepsilon_F = \frac{h^2}{8m_n}\left(\frac{3N}{\pi V}\right)^{2/3}, \quad M_{\rm NS} = Nm_n.$$

Following the same procedure as for white dwarfs, the kinetic energy is

$$U_k = \frac{\beta_k'}{R^2},$$

with

$$\beta_k' = \frac{3}{40}\left(\frac{9}{4\pi^2}\right)^{2/3}\frac{h^2}{m_n}\left(\frac{M_{\rm NS}}{m_n}\right)^{5/3}.$$

The gravitational potential energy is

$$U_p = -\frac{\alpha_p'}{R}, \quad \alpha_p' = \frac{3}{5}GM_{\rm NS}^2.$$

The virial theorem $2U_k + U_p = 0$ gives the equilibrium radius:

$$R_{\rm NS} = \frac{2\beta'_k}{\alpha'_p} = 0.093 \frac{h^2}{G m_n^{8/3} M_{\rm NS}^{1/3}}. \tag{48}$$

In the relativistic limit, the Tolman–Oppenheimer–Volkoff (TOV) limit for non-rotating neutron stars is approximately,

$$M_{\rm NS} \leq 3 M_\odot. \tag{49}$$

## 6.4 Debye Theory of Solid

Phonons obey Bose-Einstein statistics with zero chemical potential ($\mu = 0$):

$$\overline{n}_p = \frac{1}{e^{\varepsilon/k_B T} - 1}.$$

### 6.4.1 Phonon Energy

Phonons(声子) are quantized lattice vibrations . The phonon heat capacity dominates at high temperatures.

For standing waves in a box of length $L$, the quantisation condition is

$$L = n \cdot \frac{\lambda}{2},$$

so the wavelength is $\lambda = 2L/n$.

The volume of the solid is

$$V = L^3.$$

The phonon energy is related to its frequency and wavelength by

$$\varepsilon_n = h\nu = \frac{hc_s}{\lambda} = \frac{hc_s n}{2L} = \frac{hc_s n}{2V^{1/3}} \tag{50}$$

where $c_s$ is the speed of sound in the solid, and $n$ is a positive integer.

### 6.4.2 Lattice Thermal Energy

For each polarisation, the number of vibrational modes with wavevector magnitude up to $k_{\rm max}$ is

$$N = \frac{1}{8} \cdot \frac{4\pi}{3} n_{\rm max}^3 = \frac{\pi}{6} n_{\rm max}^3.$$

Here $N$ is the number of modes per polarisation; the total number of vibrational modes across all three polarisations is $3N$.

The Debye temperature is defined as the maximum phonon energy divided by $k_B$. Using $\varepsilon_{\rm max} = hc_s n_{\rm max}/(2V^{1/3})$ and $n_{\rm max}^3 = 6N/\pi$, we have:

$$T_D = \frac{\varepsilon_{\rm max}}{k_B} = \frac{hc_s}{2k_B} \left(\frac{6N}{\pi V}\right)^{1/3}. \tag{51}$$

The internal energy is obtained by summing over all phonon modes. There are three polarisations: two transverse and one longitudinal. Converting the sum to an integral over the positive octant:

$$\begin{aligned} U &= 3 \sum_{n_x \geq 1} \sum_{n_y \geq 1} \sum_{n_z \geq 1} \varepsilon_n \, \overline{n}_p(\varepsilon_n) \\ &= 3 \cdot \frac{1}{8} \int_0^{n_{\max}} \frac{\varepsilon_n}{e^{\varepsilon_n / k_B T} - 1} \cdot 4\pi n^2 \, dn. \end{aligned}$$

We introduce the dimensionless variable

$$x = \frac{\varepsilon_n}{k_B T} = \frac{T_D}{T} \cdot \frac{n}{n_{\max}}.$$

With this substitution, the Debye internal energy is:

$$U = 9 N k_B \frac{T^4}{T_D^3} \int_0^{T_D/T} \frac{x^3}{e^x - 1} \, dx. \tag{52}$$

### 6.4.3 Heat Capacity

The heat capacity is obtained by differentiating the internal energy with respect to temperature.

**a) High-temperature limit** ($T \gg T_D$): In this limit, $x \ll 1$, so we can approximate $e^x \approx 1 + x$. The integral reduces to

$$\int_0^{T_D/T} \frac{x^3}{e^x - 1} \, dx \approx \int_0^{T_D/T} x^2 \, dx = \frac{1}{3} \left( \frac{T_D}{T} \right)^3.$$

Substituting this into the expression for $U$:

$$U = 9 N k_B \frac{T^4}{T_D^3} \cdot \frac{1}{3} \left( \frac{T_D}{T} \right)^3 = 3 N k_B T.$$

This corresponds to 6 degrees of freedom per atom (3 kinetic + 3 potential). The heat capacity in the high-temperature limit:

$$C_V = \left( \frac{\partial U}{\partial T} \right)_V = 3 N k_B \quad (T \gg T_D). \tag{53}$$

This is the Dulong–Petit law.

**b) Low-temperature limit** ($T \ll T_D$): In this limit, the upper limit of the integral extends to infinity ($T_D/T \to \infty$). The integral becomes

$$\int_0^{\infty} \frac{x^3}{e^x - 1} \, dx = \frac{\pi^4}{15}.$$

Thus

$$U = 9 N k_B \frac{T^4}{T_D^3} \cdot \frac{\pi^4}{15} = \frac{3\pi^4}{5} N k_B \frac{T^4}{T_D^3}.$$

Differentiating with respect to temperature:

$$C_V = \left( \frac{\partial U}{\partial T} \right)_V = \frac{12 \pi^4}{5} N k_B \left( \frac{T}{T_D} \right)^3. \tag{54}$$

This is the Debye $T^3$ law, which holds for most insulating solids at low temperatures.

For metals, the conduction electrons contribute an additional linear term in temperature:

$$C_V = \frac{\pi^2 N k_B^2}{2\varepsilon_F} T + \frac{12\pi^4 N k_B}{5T_D^3} T^3 \quad \text{(Metals)}.$$

**c) Intermediate temperatures**: For intermediate temperatures, the full integral must be evaluated numerically:

$$C_V = 9Nk_B \left(\frac{T}{T_D}\right)^3 \int_0^{T_D/T} \frac{x^4 e^x}{(e^x - 1)^2}\, dx. \tag{55}$$

## 6.5 Bose-Einstein Condensation (BEC)

For a gas of non-interacting spin-0 bosons, the occupation number is given by the Bose-Einstein distribution:

$$\overline{n}_{\text{BE}} = \frac{1}{e^{(\varepsilon-\mu)/k_B T} - 1}.$$

### 6.5.1 Single Particle Energy

For a particle in a cubic box of volume $V = L^3$, the standing wave condition gives

$$L = \frac{n\lambda}{2},$$

so the wavevector is

$$k = \frac{2\pi}{\lambda} = \frac{n\pi}{L}.$$

The momentum and de Broglie wavelength are related by

$$p = \frac{h}{\lambda} = \frac{hk}{2\pi} = \frac{hn}{2L} = \frac{hn}{2V^{1/3}}$$

Thus the single-particle energy is

$$\varepsilon_n = \frac{p^2}{2m} = \frac{h^2 k^2}{8\pi^2 m} = \frac{h^2 n^2}{8mV^{2/3}}.$$

For spin-0 bosons such as ground-state $^4$He, there is only one spin state. The density of states in $k$-space (positive octant only) is

$$g(k)\, dk = 1 \cdot \frac{1}{8} \cdot \frac{4\pi k^2 dk}{(\pi/L)^3} = \frac{Vk^2}{2\pi^2} dk.$$

Converting to energy space using $\varepsilon_n = h^2 k^2 / 8\pi^2 m$:

$$g(\varepsilon_n) = g(k)\frac{dk}{d\varepsilon_n} = \frac{2\pi V}{h^3}(2m)^{3/2}\sqrt{\varepsilon_n}. \tag{56}$$

### 6.5.2 Ground State Population

The quantum numbers satisfy

$$n^2 = n_x^2 + n_y^2 + n_z^2.$$

The ground state corresponds to $n_x = n_y = n_z = 1$, so its energy is

$$\varepsilon_0 = \frac{h^2}{8mV^{2/3}}(1^2 + 1^2 + 1^2) = \frac{3h^2}{8mL^2}.$$

The occupation number of the ground state is therefore

$$\overline{n}_{\mathrm{BE}_0} = \frac{1}{e^{(\varepsilon_0 - \mu)/k_B T} - 1}.$$

As $T \to 0$, the chemical potential approaches the ground state energy from below:

$$\lim_{T \to 0} \mu = \varepsilon_0.$$

Consequently, the denominator tends to zero, so the ground-state occupation diverges:

$$\overline{n}_{\mathrm{BE}_0} \to \infty \quad \text{as } T \to 0.$$

This indicates macroscopic occupation of the ground state, i.e. Bose-Einstein condensation.

### 6.5.3 Critical Condensation Temperature

At temperatures $T \geq T_c$, all particles are in excited states, so the ground-state occupation is negligible. At the critical point,

$$\lim_{T \to T_c} N_0 = 0, \qquad \lim_{T \to T_c} \mu = 0.$$

The total number of particles at $T = T_c$ is obtained by integrating the density of states times the Bose-Einstein distribution:

$$\begin{aligned} N &= \lim_{T \to T_c} \int_0^\infty g(\varepsilon) \frac{1}{e^{\varepsilon / k_B T_c} - 1} \, d\varepsilon \\ &= \frac{2\pi V}{h^3} (2m)^{3/2} \int_0^\infty \frac{\sqrt{\varepsilon}}{e^{\varepsilon / k_B T_c} - 1} \, d\varepsilon. \end{aligned}$$

Introducing the dimensionless variable $x = \varepsilon / (k_B T_c)$ gives

$$N = \left( \frac{2\pi m k_B T_c}{h^2} \right)^{3/2} V \left[ \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{x}}{e^x - 1} \, dx \right].$$

For Riemann zeta function,

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx.$$

Gamma function,

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$$

And $\Gamma(3/2) = \sqrt{\pi}/2$. Thus

$$\frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{x}}{e^x - 1}\, dx = \frac{1}{\Gamma(3/2)} \int_0^\infty \frac{\sqrt{x}}{e^x - 1}\, dx = \zeta(3/2) \approx 2.612.$$

Therefore,

$$N = 2.612 \left( \frac{2\pi m k_B T_c}{h^2} \right)^{3/2} V.$$

Thus the critical temperature $T_c$ is:

$$k_B T_c = 0.527 \frac{h^2}{2\pi m} \left( \frac{N}{V} \right)^{2/3}. \tag{57}$$

For $T < T_c$, a macroscopic fraction of particles occupies the ground state. The number of particles remaining in excited states scales as

$$N_{\text{excited}} = N \left( \frac{T}{T_c} \right)^{3/2}, \quad (T < T_c).$$

The condensate fraction (the number of particles in the ground state) is therefore

$$N_0 = N - N_{\text{excited}} = N \left[ 1 - \left( \frac{T}{T_c} \right)^{3/2} \right], \quad (T < T_c). \tag{58}$$